



<Cipher/>

// COMPETITIVE PROGRAMMING

NEWCOMERS ROADMAP

Competitive Programming for Beginners

8 WEEKS

ROADMAP OVERVIEW

NEWCOMERS

ONE STAGE · EIGHT MAIN TOPICS

01

Introduction

02

Conditions

03

Loops

04

Arrays

05

Strings

06

Math & N.T.

07

Functions

08

Recursion

GENERAL PRACTICE

Easy · Medium · Hard

CODEFORCES

Div. 4 → Div. 3

PHASE 1

Data & search fundamentals

PHASE 2

Graphs, DP & backtracking

INTRO && CONDITIONS

Basic Data Types	standard input/output 1 s, 256 MB			
Simple Calculator	standard input/output 1 s, 256 MB			
Difference	standard input/output 1 s, 256 MB			
Area of a Circle	standard input/output 1 s, 256 MB			
Digits Summation	standard input/output 0.25 s, 64 MB			
Summation from 1 to N	standard input/output 0.25 s, 256 MB			
Two numbers	standard input/output 1 s, 256 MB			
Welcome for you with Conditions	standard input/output 1 s, 64 MB			
Multiples	standard input/output 1 s, 256 MB			
Max and Min	standard input/output 0.25 s, 64 MB			

SHEET URL:

<https://codeforces.com/group/MWSDmqGsZm/contest/219158>

WHAT YOU'LL LEARN

Learn the basic building blocks of C++ programming: writing simple programs, declaring variables, and using input, output, and operators.

Understand how programs make decisions using conditions, and how Boolean expressions control the flow of a program.

TOPICS

- C++ Basics
- Variables & Data Types
- Input / Output
- if / else
- Boolean expressions

WEEKLY GOAL

Understand the basic syntax and building blocks needed to start solving simple programming problems in C++.

Be able to write programs that make decisions based on different conditions.

LOOPS

Even Numbers	standard input/output 1 s, 256 MB	
Even, Odd, Positive and Negative	standard input/output 1 s, 256 MB	
Fixed Password	standard input/output 1 s, 256 MB	
Max	standard input/output 1 s, 256 MB	
Multiplication table	standard input/output 1 s, 64 MB	
Factorial	standard input/output 2 s, 64 MB	
One Prime	standard input/output 3 s, 64 MB	
Palindrome	standard input/output 1 s, 256 MB	
Primes from 1 to n	standard input/output 3 s, 256 MB	
Divisors	standard input/output 1 s, 256 MB	

SHEET URL:

<https://codeforces.com/group/MWSDmqGsZm/contest/219432>

WHAT YOU'LL LEARN

Understand repetition and iteration: executing code multiple times, choosing the right loop type, and controlling execution.

TOPICS

- for
- while
- do while
- break / continue

WEEKLY GOAL

Be comfortable using loops to solve repetitive and counting-based problems.

ARRAYS

Searching	standard input/output 2 s, 64 MB	
Replacement	standard input/output 1 s, 256 MB	
Positions in array	standard input/output 1 s, 256 MB	
Lowest Number	standard input/output 1 s, 256 MB	
Reversing	standard input/output 1 s, 64 MB	
Palindrome Array	standard input/output 1 s, 256 MB	
Sorting	standard input/output 1 s, 64 MB	
Smallest Pair	standard input/output 1 s, 256 MB	
Lucky Array	standard input/output 1 s, 256 MB	
Sum Digits	standard input/output 2 s, 256 MB	

SHEET URL:

<https://codeforces.com/group/MWSDmqGsZm/contest/219774>

WHAT YOU'LL LEARN

Understand arrays as a way to store and work with multiple values, including one-dimensional and basic two-dimensional arrays.

TOPICS

- 1D Arrays
- 2D Arrays

WEEKLY GOAL

Be able to store, access, and process collections of values using arrays.

STRINGS

Let's use Getline	standard input/output 1 s, 64 MB	
Compare	standard input/output 1 s, 256 MB	
Strings	standard input/output 2 s, 64 MB	
Count	standard input/output 2 s, 64 MB	
Way Too Long Words	standard input/output 1 s, 256 MB	
Conversion	standard input/output 2 s, 64 MB	
Good or Bad	standard input/output 2 s, 64 MB	
Palindrome	standard input/output 1 s, 256 MB	
Count Letters	standard input/output 2 s, 64 MB	
I Love strings	standard input/output 2 s, 64 MB	

SHEET URL:

<https://codeforces.com/group/MWSDmqGsZm/contest/219856>

WHAT YOU'LL LEARN

Learn how to work with text: the string class, accessing characters, and performing basic string operations.

TOPICS

- string class
- Working with characters

WEEKLY GOAL

Be comfortable solving basic problems involving strings and characters.

FUNCTIONS

Print	standard input/output 1 s, 64 MB	
Wonderful Number	standard input/output 1 s, 64 MB	
Prime Function	standard input/output 2 s, 256 MB	
Swap	standard input/output 1 s, 64 MB	
Equation	standard input/output 1 s, 256 MB	
Max and MIN	standard input/output 1 s, 64 MB	
N Times	standard input/output 1 s, 256 MB	
Swapping With Matrix	standard input/output 1 s, 256 MB	
Average	standard input/output 1 s, 256 MB	
Shift Right	standard input/output 1 s, 256 MB	

SHEET URL:

<https://codeforces.com/group/MWSDmqGsZm/contest/223205>

WHAT YOU'LL LEARN

Understand how functions organize and reuse code, including parameters, pass by value vs. reference, and passing arrays.

TOPICS

- Parameters
- Pass by value / reference
- Passing arrays

WEEKLY GOAL

Be able to write reusable functions and understand how data is passed between functions.

MATH & Geometry

Prime checking	standard input/output 4 s, 256 MB	
Sum of Range	standard input/output 0.5 s, 256 MB	
Xor	standard input/output 1 s, 256 MB	
Maximum Distinct Numbers	standard input/output 1 s, 64 MB	
Multiplication of Matrices	standard input/output 1 s, 256 MB	
Summation of its divisors	standard input/output 1 s, 256 MB	
GCD	standard input/output 1 s, 256 MB	
Divisability	standard input/output 0.25 s, 64 MB	
Prime Factors	standard input/output 1 s, 256 MB	
Combination and Permutation	standard input/output 1 s, 256 MB	

SHEET URL:

<https://codeforces.com/group/MWSDmqGsZm/contest/223338>

WHAT YOU'LL LEARN

Get familiar with the math concepts used often in beginner Competitive Programming, from number properties to combinatorics.

TOPICS

- cmath
- Number Theory basics
- GCD
- Prime numbers
- Decimal / Binary
- Pascal Triangle
- Permutations & Combinations

WEEKLY GOAL

Build the mathematical foundation needed for solving basic Competitive Programming problems.

RECURSION

Print from 1 to N	standard input/output 1 s, 64 MB
Print from N to 1	standard input/output 1 s, 64 MB
Print Digits using Recursion	standard input/output 1 s, 256 MB
Base Conversion	standard input/output 1 s, 256 MB
Print Even Indices	standard input/output 1 s, 256 MB
Pyramid	standard input/output 1 s, 256 MB
Inverted Pyramid	standard input/output 1 s, 256 MB
Count Vowels	standard input/output 1 s, 256 MB
Factorial	standard input/output 1 s, 64 MB
Max Number	standard input/output 1 s, 64 MB

SHEET URL:

<https://codeforces.com/group/MWSDmqGsZm/contest/223339>


WHAT YOU'LL LEARN

Understand recursion: a function calling itself to break a problem down into smaller versions of the same problem.



TOPICS

- Recursion



WEEKLY GOAL

Understand the basic idea behind recursion and be able to trace and implement simple recursive solutions.

GENERAL PRACTICE

Extra Practice for Newcomers

After completing the eight Newcomers topics, solve additional problems to reinforce what you've learned and build real problem-solving experience, before moving on to the next stage.

● EASY

Additional practice for reinforcing the fundamentals.

GENERAL — EASY:

<https://codeforces.com/group/MWSDmqGsZm/contest/223206>

● MEDIUM

Problems that combine multiple concepts from the Newcomers topics.

GENERAL — MEDIUM:

<https://codeforces.com/group/MWSDmqGsZm/contest/223207>

● HARD

More challenging problems requiring deeper use of the concepts learned.

GENERAL — HARD:

<https://codeforces.com/group/MWSDmqGsZm/contest/223340>

CODEFORCES

DIV. 4



DIV. 3

These beginner-friendly contests provide practical experience applying the concepts learned during the Newcomers stage, helping students transition from guided practice to real contest problem solving.

NEWCOMERS



GENERAL PRACTICE



CODEFORCES



NEXT STEPS

NEXT STEPS

You've completed the Newcomers stage and its General Practice. From here, the roadmap moves beyond guided weekly topics and into two broader phases of study.

PHASE 1



PHASE 2

PHASE 1

Data & Search Fundamentals

● SHORT DESCRIPTION

Phase 1 moves from writing basic programs to reasoning about data efficiently. Building on the C++ fundamentals from Newcomers, learners begin applying techniques for handling larger inputs, faster lookups, and more structured problem patterns.

● WHAT YOU'LL GAIN

Learners leave Phase 1 able to choose the right data structure or technique for a problem's size and constraints, and to write cleaner, more efficient C++ using the STL.

● MAIN AREAS

- **Sorting & Searching**
- **Prefix Sums & Frequency Arrays**
- **STL (vector, set, map, pair)**
- **Two Pointers**
- **Binary Search**
- **Introductory Greedy**

PHASE 2

Algorithms & Problem Modeling

● SHORT DESCRIPTION

Phase 2 builds on the data-handling skills from Phase 1 and introduces the algorithmic techniques that appear throughout ICPC-style contests — exploring graphs, breaking problems into subproblems, and reasoning systematically about state and choices.

● WHAT YOU'LL GAIN

Learners finish Phase 2 comfortable modeling problems as graphs or recurrences, and ready to tackle harder Div. 3 / Div. 2 and mid-level ICPC regional-style problems.

● MAIN AREAS

● Graphs (BFS / DFS)

● Trees

● Dynamic Programming

● Backtracking

● Divide & Conquer

● Advanced Greedy

STUDY RESOURCES

A clean reference index for the Newcomers stage

NEWCOMERS / CODEFORCES

01 — Introduction

<https://codeforces.com/group/MWSDmqGsZm>

02 — Conditions

<https://codeforces.com/group/MWSDmqGsZm/contest/219158>

03 — Loops

<https://codeforces.com/group/MWSDmqGsZm/contest/219432>

04 — Arrays

<https://codeforces.com/group/MWSDmqGsZm/contest/219774>

05 — Strings

<https://codeforces.com/group/MWSDmqGsZm/contest/219856>

06 — Math & Number Theory Basics

<https://codeforces.com/group/MWSDmqGsZm/contest/223338>

07 — Functions

<https://codeforces.com/group/MWSDmqGsZm/contest/223205>

08 — Recursion

<https://codeforces.com/group/MWSDmqGsZm/contest/223339>

VIDEO MATERIALS

C++ Basics

Video 1 https://youtu.be/Fr6wJ5_Hok0?si=gxsvobcmsSMOxrT2

Video 2 <https://youtu.be/H7pUHZrgscY?si=iK5NzcMvfQsrDLZc>

Video 3 <https://youtu.be/-ZRhEfIMx7RU?si=ezvGxh7AYSjkfbY8>

Conditions

Open video <https://www.youtube.com/watch?v=02ZmSVsoAV4>

Loops

Video 1 <https://youtu.be/OGweo6b6k0c?si=88lcYCBbMA8n0HXB>

Video 2 <https://youtu.be/TLiasndFVVc?si=DtWp5vYmXS94JXnw>

Arrays

Open video <https://youtu.be/-Xfx53vVvR0?si=sAEpi7zzqh-kP3io>

Strings

Open video <https://www.youtube.com/watch?v=HOB-Ey3tEqY>

Functions

Open video <https://www.youtube.com/watch?v=dQZZg8okYKg&t=8581s>

Recursion

Video 1 <https://www.youtube.com/watch?v=sXuxl4TBGfo>

Video 2 <https://www.youtube.com/watch?v=-W0Ftkel5ek>